

## DUKE FIELD, FL, USA

WMO: 722246

Lat: 30.650N

Lon: 86.517W

Elev: 58

StdP: 100.63

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 03844

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |             |            |            | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|------------|-------------|------------|------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |            | 99%         |            |            | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP          | HR         | MCDB       | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-3.8 | (c)<br>-2.1 | (d)<br>-12.3                  | (e)<br>1.3 | (f)<br>2.2 | (g)<br>-9.8 | (h)<br>1.6 | (i)<br>4.1 | (j)<br>10.6           | (k)<br>15.4 | (l)<br>9.2 | (m)<br>12.1 | (n)<br>1.7            | (o)<br>340 | (p)<br>0.378 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.1                   | 35.1            | 24.5 | 34.0 | 24.5 | 32.9 | 24.4 | 26.9                | 31.4 | 26.4 | 30.8 | 25.8 | 30.1 | 3.4                  | 330  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.0                            | 21.5 | 28.5 | 25.2 | 20.5 | 28.0 | 24.9 | 20.1 | 27.8 | 84.7          | 31.5 | 82.3 | 30.8 | 80.1 | 30.2 | 29.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 8.5               | 7.6  | 6.8 | -7.4                       | 36.7 | 2.0                | 1.3 | -8.8                                               | 37.6 | -10.0      | 38.3 | -11.2      | 39.0 | -12.6      | 39.9 |  |  |
|                   |      |     | -8.1                       | 28.0 | 2.1                | 0.6 | -9.6                                               | 28.4 | -10.8      | 28.8 | -12.0      | 29.1 | -13.5      | 29.6 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 19.6   | 10.0  | 12.5  | 15.6  | 19.0  | 23.4  | 26.8  | 27.4  | 27.3  | 25.5  | 20.3  | 14.2  | 12.4  |
|                                                                                  | DBStd       | 7.32   | 5.84  | 5.23  | 4.69  | 3.25  | 2.79  | 1.79  | 1.49  | 1.56  | 2.29  | 4.16  | 4.56  | 5.37  |
|                                                                                  | HDD10.0     | 169    | 76    | 33    | 9     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 13    | 37    |
|                                                                                  | HDD18.3     | 941    | 264   | 172   | 110   | 31    | 4     | 0     | 0     | 0     | 0     | 29    | 138   | 194   |
|                                                                                  | CDD10.0     | 3658   | 74    | 102   | 182   | 271   | 414   | 503   | 539   | 535   | 466   | 319   | 139   | 112   |
|                                                                                  | CDD18.3     | 1389   | 5     | 8     | 24    | 51    | 160   | 253   | 281   | 277   | 217   | 89    | 14    | 11    |
|                                                                                  | CDH23.3     | 13067  | 4     | 24    | 149   | 462   | 1528  | 2572  | 2791  | 2646  | 1926  | 841   | 101   | 24    |
|                                                                                  | CDH26.7     | 5313   | 0     | 1     | 17    | 77    | 586   | 1182  | 1231  | 1133  | 821   | 258   | 8     | 0     |
| Wind                                                                             | WSAvg       | 2.7    | 3.1   | 3.2   | 3.3   | 3.2   | 2.8   | 2.4   | 2.2   | 2.0   | 2.1   | 2.4   | 2.5   | 2.9   |
| Precipitation                                                                    | PrecAvg     | 1505   | 108   | 116   | 134   | 129   | 85    | 136   | 178   | 157   | 138   | 94    | 105   | 123   |
|                                                                                  | PrecMax     | 2093   | 419   | 211   | 325   | 418   | 294   | 306   | 401   | 249   | 645   | 304   | 303   | 290   |
|                                                                                  | PrecMin     | 975    | 6     | 34    | 15    | 26    | 11    | 36    | 66    | 46    | 35    | 6     | 10    | 49    |
|                                                                                  | PrecStd     | 319    | 79    | 51    | 84    | 82    | 71    | 72    | 82    | 51    | 119   | 86    | 82    | 57    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 23.9  | 25.8  | 28.3  | 30.0  | 34.9  | 37.2  | 35.9  | 36.1  | 35.5  | 32.5  | 27.8  | 25.2  |
|                                                                                  |             | MCWB   | 19.4  | 19.2  | 19.0  | 20.6  | 22.5  | 24.1  | 25.1  | 24.7  | 23.5  | 22.9  | 20.4  | 20.5  |
|                                                                                  | 2%          | DB     | 22.5  | 23.9  | 26.9  | 28.6  | 32.7  | 34.9  | 34.5  | 34.3  | 33.7  | 31.0  | 25.8  | 24.0  |
|                                                                                  |             | MCWB   | 19.3  | 18.4  | 18.7  | 19.5  | 22.2  | 24.6  | 25.2  | 25.3  | 23.7  | 22.2  | 19.6  | 20.1  |
|                                                                                  | 5%          | DB     | 21.3  | 22.3  | 25.0  | 27.5  | 31.2  | 33.7  | 33.2  | 33.2  | 32.4  | 29.6  | 24.3  | 22.6  |
|                                                                                  |             | MCWB   | 18.6  | 18.3  | 18.2  | 19.0  | 21.9  | 24.3  | 25.0  | 25.1  | 24.0  | 22.0  | 18.5  | 20.0  |
|                                                                                  | 10%         | DB     | 19.8  | 20.9  | 23.0  | 26.2  | 30.0  | 32.4  | 32.3  | 32.2  | 31.2  | 27.9  | 22.8  | 21.1  |
|                                                                                  |             | MCWB   | 17.5  | 18.3  | 17.2  | 18.9  | 21.8  | 24.2  | 24.8  | 25.0  | 24.1  | 20.8  | 18.2  | 19.1  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 21.5  | 21.6  | 22.0  | 23.8  | 25.2  | 27.0  | 27.3  | 27.5  | 27.9  | 25.9  | 23.6  | 23.0  |
|                                                                                  |             | MCDB   | 22.4  | 22.9  | 24.7  | 26.2  | 29.8  | 32.1  | 32.4  | 32.5  | 31.2  | 28.4  | 25.1  | 23.8  |
|                                                                                  | 2%          | WB     | 20.6  | 20.7  | 21.1  | 22.7  | 24.2  | 26.3  | 26.6  | 26.9  | 26.6  | 24.9  | 22.0  | 21.8  |
|                                                                                  |             | MCDB   | 21.6  | 22.0  | 23.4  | 24.9  | 28.4  | 31.1  | 31.6  | 31.5  | 29.5  | 27.7  | 23.7  | 22.8  |
|                                                                                  | 5%          | WB     | 19.4  | 19.9  | 20.3  | 22.0  | 23.7  | 25.7  | 26.0  | 26.4  | 25.5  | 24.1  | 20.5  | 20.7  |
|                                                                                  |             | MCDB   | 20.5  | 21.2  | 22.6  | 24.8  | 28.0  | 30.6  | 30.8  | 30.6  | 29.1  | 27.2  | 23.1  | 21.9  |
|                                                                                  | 10%         | WB     | 17.6  | 18.8  | 19.2  | 21.0  | 23.1  | 25.2  | 25.6  | 25.8  | 24.9  | 23.0  | 19.1  | 19.3  |
|                                                                                  |             | MCDB   | 18.9  | 20.5  | 21.9  | 23.9  | 27.4  | 30.0  | 30.1  | 29.8  | 28.7  | 26.2  | 22.0  | 20.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 12.2  | 12.4  | 12.9  | 13.1  | 12.4  | 10.9  | 10.1  | 9.8   | 11.1  | 13.5  | 14.5  | 11.5  |
|                                                                                  |             | MCDBR  | 10.6  | 11.5  | 13.0  | 14.2  | 14.3  | 12.8  | 11.4  | 11.5  | 12.7  | 13.6  | 14.3  | 10.8  |
|                                                                                  |             | MCWBR  | 7.9   | 7.4   | 6.5   | 6.5   | 5.5   | 4.3   | 4.0   | 4.2   | 4.6   | 6.1   | 8.7   | 7.9   |
|                                                                                  | 5% WB       | MCDBR  | 8.9   | 9.3   | 9.9   | 9.2   | 11.2  | 11.1  | 10.0  | 9.6   | 10.0  | 10.0  | 11.2  | 8.9   |
|                                                                                  |             | MCWBR  | 7.9   | 7.6   | 6.1   | 5.6   | 5.1   | 4.5   | 4.0   | 4.0   | 4.7   | 5.8   | 8.4   | 8.0   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.329 | 0.343 | 0.365 | 0.393 | 0.433 | 0.465 | 0.507 | 0.497 | 0.435 | 0.380 | 0.352 | 0.337 |
|                                                                                  | taud        |        | 2.543 | 2.505 | 2.438 | 2.368 | 2.287 | 2.253 | 2.153 | 2.176 | 2.324 | 2.456 | 2.475 | 2.529 |
|                                                                                  | Ebn at Noon |        | 905   | 921   | 919   | 900   | 861   | 829   | 793   | 798   | 838   | 868   | 867   | 875   |
|                                                                                  | Edh at Noon |        | 86    | 97    | 111   | 123   | 134   | 138   | 153   | 147   | 123   | 101   | 91    | 83    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 3.09  | 3.69  | 5.00  | 6.03  | 6.66  | 6.29  | 5.96  | 5.53  | 5.14  | 4.54  | 3.50  | 2.71  |
|                                                                                  | RadStd      |        | 0.22  | 0.47  | 0.39  | 0.48  | 0.39  | 0.52  | 0.37  | 0.41  | 0.32  | 0.45  | 0.31  | 0.28  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (41 neighbors) |  | +0.45 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +154    | +97     |

Nomenclature: See separate page